2-DOF Helicopter Flight Dynamics Modeling and LQR Control

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Project Overview

Figure 1: Quanser Aero 2 two-degree-of-freedom helicopter platform used for flight dynamics modeling and control design.

This project develops a flight dynamics model and control system for the Quanser Aero 2 two-degree-of-freedom helicopter platform. The objective is to design a controller capable of tracking multiple reference trajectories while maintaining system stability.

The helicopter dynamics are described using a state-space model with pitch (θ) and yaw (ψ) as the primary outputs.

System Model

The system state vector is defined as

$$x = \begin{bmatrix} \theta \\ \psi \\ \dot{\theta} \\ \dot{\psi} \end{bmatrix}$$

The linearized dynamics are

$$\dot{x} = Ax + Bu$$

$$y = Cx$$

where

$$A = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ -0.3208 & 0 & 0.0858 & 0 \\ 0 & 0 & 0 & -0.0806 \end{bmatrix}$$

$$B = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0.0232 & 0.0099 \\ -0.0224 & 0.0429 \end{bmatrix}$$

$$C = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}$$

Integral State Augmentation

To achieve reference tracking, two integral states are introduced:

$$\dot{x}_5 = \theta - r_1$$

$$\dot{x}_6 = \psi - r_2$$

The augmented state vector becomes

$$x_a = \begin{bmatrix} x \\ x_5 \\ x_6 \end{bmatrix}$$

The augmented system is

$$\dot{x}_a = \begin{bmatrix} A & 0 \\ C & 0 \end{bmatrix} x_a + \begin{bmatrix} B \\ 0 \end{bmatrix} u$$

LQR Controller Design

The optimal control law is obtained using Linear Quadratic Regulation (LQR) by minimizing

$$J = \int_0^\infty (x_a^T Q x_a + u^T R u) dt$$

The weighting matrices used were

$$Q = \begin{bmatrix} 300 & 0 & 0 & 0 & 0 & 0 \\ 0 & 300 & 0 & 0 & 0 & 0 \\ 0 & 0 & 30 & 0 & 0 & 0 \\ 0 & 0 & 0 & 30 & 0 & 0 \\ 0 & 0 & 0 & 0 & 2000 & 0 \\ 0 & 0 & 0 & 0 & 0 & 2000 \end{bmatrix}$$

$$R = \begin{bmatrix} 0.01 & 0 \\ 0 & 0.01 \end{bmatrix}$$

The resulting control law is

$$u = -K\hat{x}_a$$

Observer Design

Since only the pitch and yaw angles are measured, the remaining states must be estimated. A Luenberger observer is therefore designed to reconstruct the full system state.

The observer dynamics are

$$\dot{\hat{x}} = A\hat{x} + Bu + L(y - C\hat{x})$$

where:

$$\hat{x} = \begin{bmatrix} \hat{\theta} \\ \hat{\psi} \\ \dot{\hat{\theta}} \\ \dot{\hat{\psi}} \end{bmatrix}$$

is the estimated state vector and L is the observer gain matrix.

The estimation error is defined as

$$e = x - \hat{x}$$

The estimation error dynamics are therefore

$$\dot{e} = (A - LC)e$$

The observer gain matrix L is chosen such that the eigenvalues of $A - LC$ lie sufficiently far in the left half of the complex plane to ensure rapid convergence of the state estimates.

Observer-Based Closed-Loop System

Combining the LQR controller with the Luenberger observer results in the observer-based feedback system.

Substituting the control law $u = -K\hat{x}_a$ and the observer dynamics into the plant equations yields the closed-loop system

$$\begin{bmatrix} \dot{x} \\ \dot{\hat{x}} \end{bmatrix} = \begin{bmatrix} A & -BK \\ LC & A - LC - BK \end{bmatrix} \begin{bmatrix} x \\ \hat{x} \end{bmatrix}$$

This structure combines optimal state feedback with state estimation, allowing the controller to stabilize the helicopter and track reference trajectories despite partial state measurements.

Simulation Results

The closed-loop system performance was evaluated using several reference trajectories. Tracking performance was analyzed through both time-domain responses and phase trajectories.

Pitch and Yaw Tracking

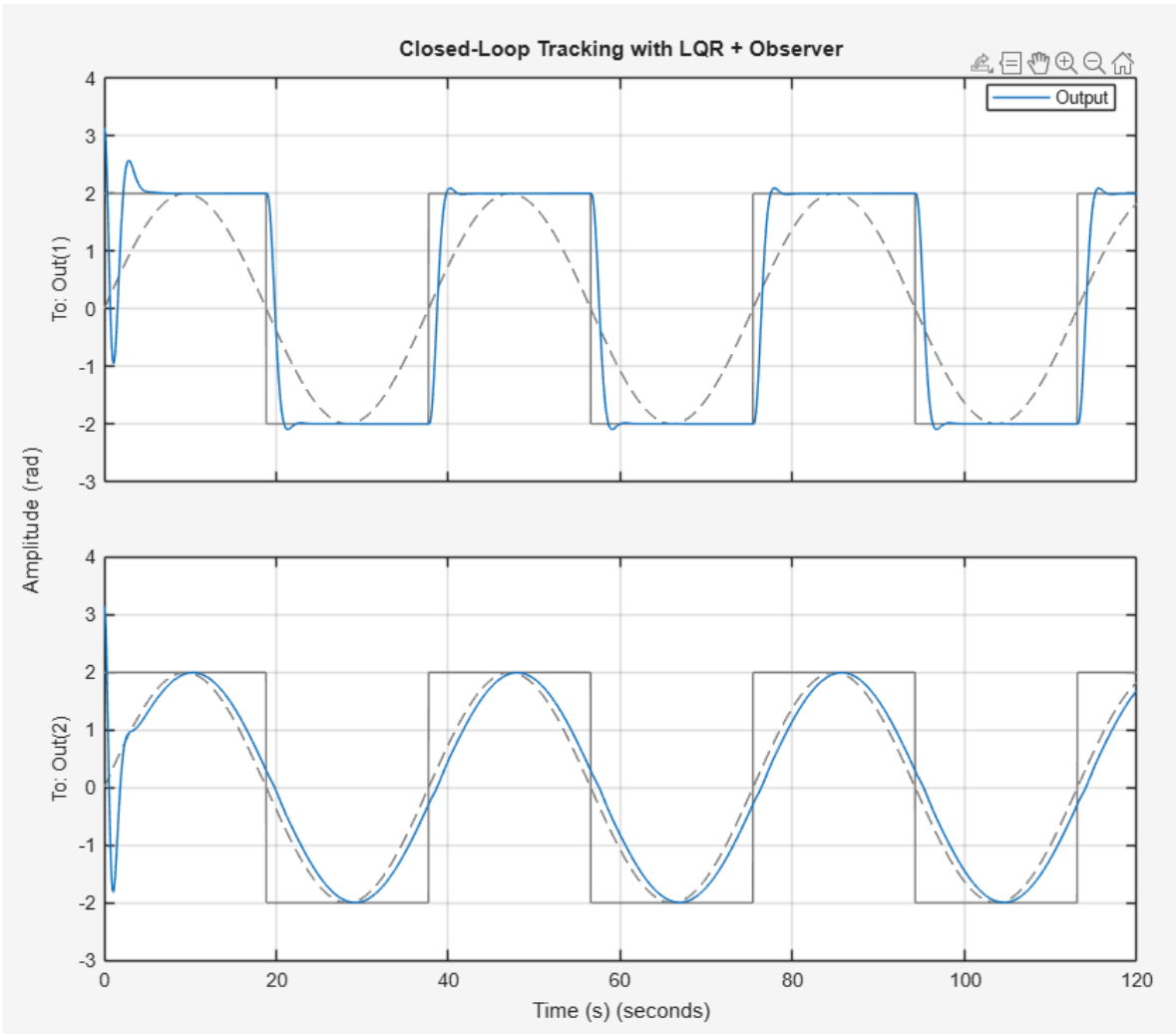
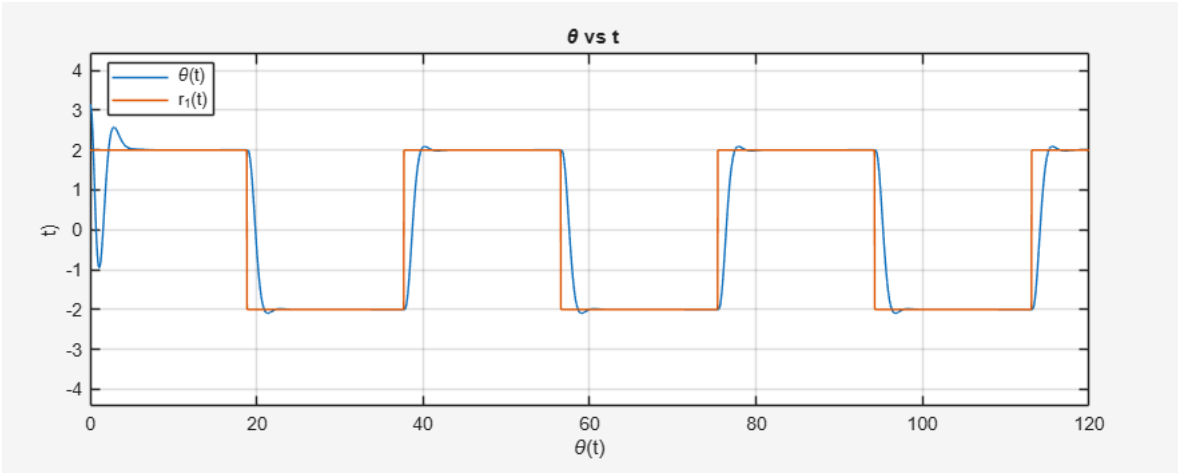


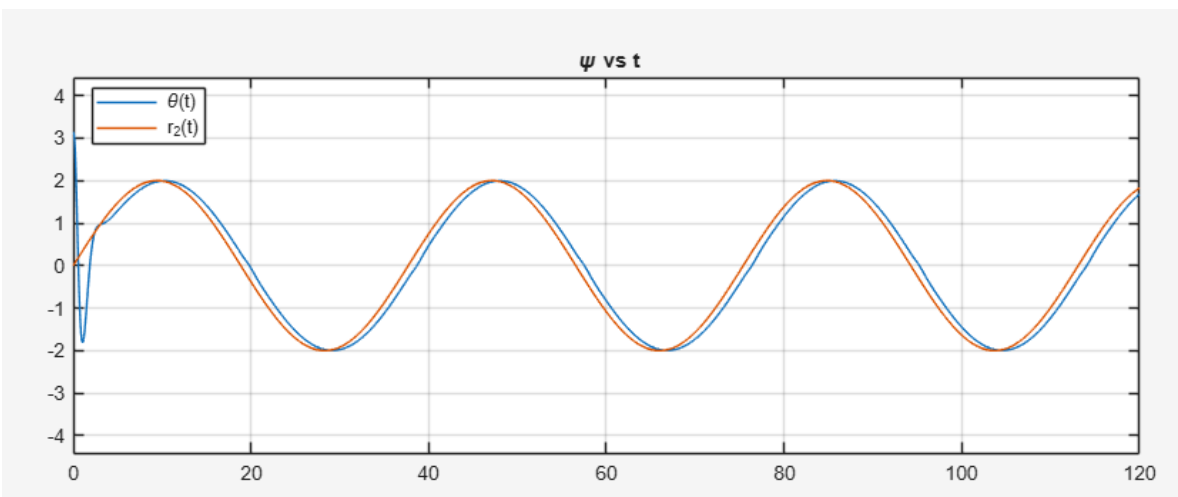
Figure 2: Closed-loop tracking of square-wave and sinusoidal references.

Square Wave Tracking



The pitch output $\theta(t)$ follows the square-wave reference input with small overshoot and rapid convergence.

Sinusoidal Tracking

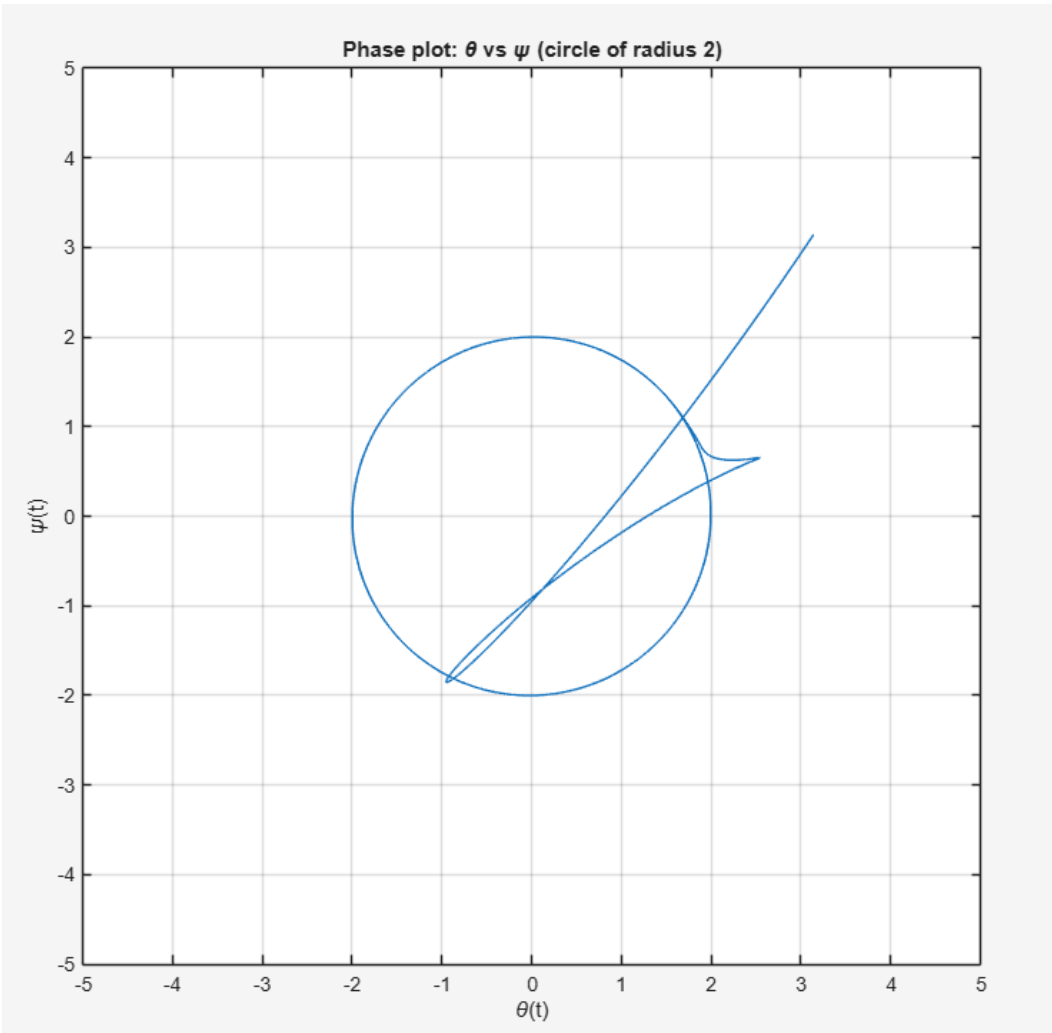


The yaw angle $\psi(t)$ closely follows the sinusoidal reference signal with smooth tracking behavior.

Circular Trajectory Tracking

Circular motion is generated using

$$r_1(t) = 2\cos(\omega t), \quad r_2(t) = 2\sin(\omega t)$$



The phase trajectory θ vs ψ converges to a circle of radius 2 in steady state.

Rotated Elliptical Trajectory

An elliptical reference trajectory is generated using

$$x(t) = a\cos(\omega t)$$

$$y(t) = b\sin(\omega t)$$

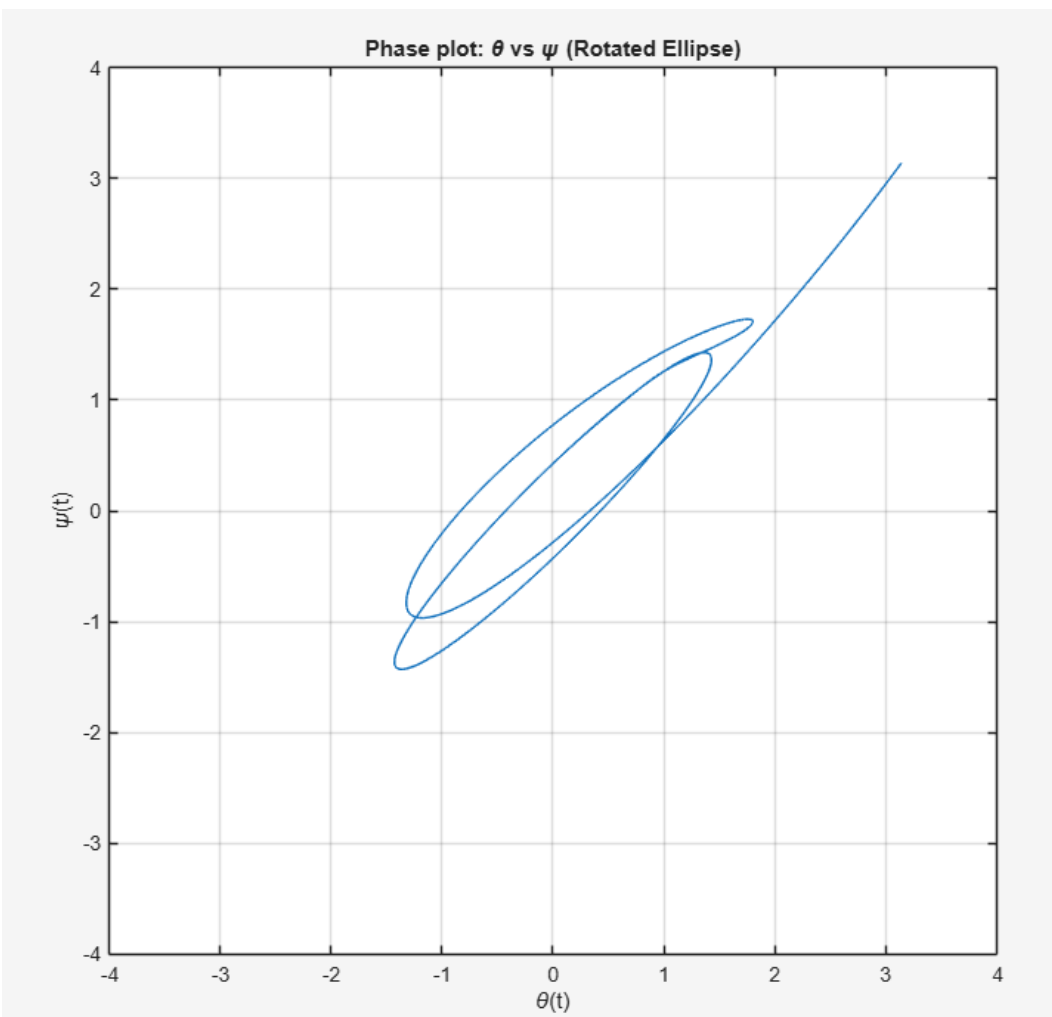
where the semi-axes are

$$a = 2, \quad b = \frac{1}{3}.$$

To produce a rotated ellipse, the reference signals are rotated by an angle $\phi = \frac{\pi}{4}$:

$$r_1(t) = x(t)\cos\phi - y(t)\sin\phi$$

$$r_2(t) = x(t)\sin\phi + y(t)\cos\phi$$



The system output converges to the desired rotated elliptical trajectory.

Tools Used

- MATLAB
- Control Systems Toolbox
- State-space modeling
- Optimal control (LQR)